CourseCompass Technical Report

Association Rule Mining for Course Recommendations

Assignment 01

1 Introduction and Problem Statement

CourseCompass recommends courses using patterns in learner enrollments. Each learner's course enrollments are treated as one basket. Apriori finds course combinations that appear often, and association rules use those combinations to suggest a course a learner has not selected.

The system also shows the rule and its support, confidence, and lift so a user can see why a course was suggested. These results describe courses taken together in the dataset. They do not show that taking one course causes a learner to take another.

2 Application Domain

The project uses online course enrollment as its application domain. A learner who enrolled in more than one course forms a multi-item basket. For example, a rule can show how often learners enrolled in The Ancient Greek Hero also enrolled in Justice: What's the Right Thing to Do? CourseCompass uses these patterns to suggest related courses and explain the evidence behind each suggestion.

3 Dataset and Preprocessing

The data comes from the HarvardX Person-Course Academic Year 2013 de-identified dataset, version 3.0. The project uses five course IDs from the dataset. The raw file contains one learner-course record per enrollment.

Measure	Value	Meaning
Enrollment records	338,223	Rows in the project dataset
Unique learners	301,609	Distinct de-identified learner IDs
Courses	5	Course IDs used by CourseCompass
Learners in 2 or more courses	30,267 (10.04%)	Baskets used for the main rule-mining results
Learners in 1 course	271,342 (89.96%)	Single-course baskets
Largest basket	5 courses	Maximum course count for one learner

The five courses and their enrollment counts are shown below. Counts refer to enrollment records, so one learner may appear in more than one course.

Course	Title	Enrollments
CS50x	Introduction to Computer Science I	169,621
ER22x	Justice: What's the Right Thing to Do?	57,406
PH207x	Health in Numbers: Quantitative Methods	41,592
PH278x	Human Health and Global Environmental Change	39,602
CB22x	The Ancient Greek Hero	30,002

Preprocessing

The preprocessing script checks the learner and course identifiers, removes rows with missing identifiers, trims extra spaces, and removes duplicate learner-course pairs. It then maps course IDs to readable course names and groups each learner's courses into a basket. The project's reported cleaned row count is unchanged at 338,223.

Most learners in the data took only one course, so they cannot contribute to a co-enrollment rule. For the main Apriori results, the project uses the 30,267 baskets containing at least two courses. The application also supports the full learner set. Results based on the multi-course group use that group as the support denominator.

Dataset source: HarvardX, HarvardX Person-Course Academic Year 2013 De-Identified Dataset, version 3.0, Harvard Dataverse, https://doi.org/10.7910/DVN/26147.

4 Association Rule Mining

Apriori starts with single courses, counts their support, and keeps those that meet the minimum support. It then joins frequent itemsets to form larger candidates. Before counting a candidate, it checks whether all of the candidate's smaller subsets are frequent. If any subset is infrequent, Apriori prunes the candidate because it cannot be frequent.

Measure	Formula	Meaning
Support	count(A and B) / N	Share of baskets containing both A and B.
Confidence	support(A and B) / support(A)	Share of A baskets that also contain B.
Lift	confidence(A → B) / support(B)	Compares the rule with B's overall frequency. Lift above 1 means B appears with A more often than its baseline rate.
Leverage	support(A and B) − support(A) × support(B)	Difference between observed and expected co-occurrence.
Conviction	(1 − support(B)) / (1 − confidence)	Measures how often A occurs without B, compared with independence.

Support for A → B is the same as support for B → A because both count baskets containing A and B. Confidence can differ because each direction uses a different antecedent. For example, the confidence of Greek Hero → Justice is 66.87%, while Justice → Greek Hero is 45.11%. Justice appears in more baskets, so it has a larger denominator when it is the antecedent.

Apriori Example

Consider five baskets: {CS, Ethics}, {CS, Stats, Ethics}, {Stats, Bio}, {CS, Stats, Bio}, and {CS, Ethics}. With minimum support 0.40, an itemset must appear in at least two baskets.

At the first level, CS appears 4 times, Ethics 3 times, Stats 3 times, and Bio 2 times. All four single items are frequent. At the second level, CS-Ethics appears 3 times, CS-Stats 2 times, and Stats-Bio 2 times. The other pairs are removed because they occur fewer than twice.

The possible triple {CS, Ethics, Stats} is pruned before its support is counted. Its subset {Ethics, Stats} was not frequent, so the Apriori property says the triple cannot be frequent either. This is the candidate generation and pruning step used by the project's Apriori engine.

5 Frequent Itemsets and Rules

Using the multi-course baskets with minimum support 0.05 and minimum confidence 0.15, the project reports 15 frequent itemsets and 25 rules. Support is the share of the 30,267 multi-course baskets containing the itemset or both sides of a rule.

Frequent Itemsets

Size	Itemset	Support	Baskets
1	Justice	0.5785	17,509
1	CS50x	0.4957	15,002
1	Environmental Health	0.4534	13,724
1	Greek Hero	0.3902	11,811
1	Health in Numbers	0.2919	8,835
2	Greek Hero + Justice	0.2609	7,898
2	Justice + Environmental Health	0.2426	7,343
2	CS50x + Justice	0.2178	6,591
2	CS50x + Health in Numbers	0.1597	4,833
2	Greek Hero + CS50x	0.1361	4,118
2	Greek Hero + Environmental Health	0.1325	4,011
2	CS50x + Environmental Health	0.1234	3,735
2	Health in Numbers + Environmental Health	0.1191	3,605
3	Greek Hero + Justice + Environmental Health	0.0753	2,279
3	Greek Hero + CS50x + Justice	0.0616	1,863

Association Rules

Rule	Support	Confidence	Lift
Greek Hero → Justice	0.2609	66.87%	1.16
Justice → Greek Hero	0.2609	45.11%	1.16
Health in Numbers → CS50x	0.1597	54.70%	1.10
Environmental Health → Justice	0.2426	53.50%	0.92
Greek Hero → CS50x	0.1361	34.87%	0.70
Greek Hero + CS50x → Justice	0.0616	45.24%	0.78

Confidence should be read with lift. For example, Environmental Health → Justice has 53.50% confidence, but its lift is 0.92. Learners in that group took Justice less often than the overall baseline rate. A high confidence value by itself does not always mean the pair has a stronger-than-baseline association.

6 Recommendation Method

When a user selects courses, CourseCompass finds rules whose antecedent is contained in the selected basket. It adds the rule's consequent as a candidate, removes any course already selected, and ranks the remaining candidates. If several rules support the same course, the system keeps the evidence for each rule and shows the strongest rule as the main explanation.

The application offers three ranking choices: highest confidence, highest lift, or a hybrid score based on confidence, lift, and the number of supporting rules. The hybrid score is max confidence × √(max lift) × [1 + 0.05 × min(rule count − 1, 4)]. For each result, the system displays the supporting rule, its metrics, and a sentence such as: “Learners who enrolled in Greek Hero also enrolled in Justice in 66.9% of baskets.” A recommendation based on a rule with lift below 1 should be treated cautiously because the consequent is less common in that antecedent group than its overall baseline rate.

7 Interactive Application

The Streamlit application includes an overview of the dataset, controls for minimum support and confidence, a frequent-itemset view, and a rules explorer with search and sorting. Users can select courses in a basket and see recommendations with their supporting rules.

The interface also includes a threshold comparison, a Smart Basket that updates as courses are added, a what-if view for removing a course, and an association graph. The charts show course frequency, basket sizes, and rule metrics. These features let users inspect how the output changes when they change the basket or mining thresholds.

8 Threshold Comparison

The project compares a conservative setting with a more exploratory one. Lowering minimum support and confidence allows more patterns and rules to pass, but it also admits weaker associations.

Result	Conservative	Exploratory
Minimum support	0.10	0.03
Minimum confidence	0.40	0.15
Frequent itemsets	13	16
Association rules	9	26
Mean rule confidence	52.6%	38.4%
Mean rule lift	1.01	0.86
General effect	Fewer, higher-confidence rules	More rules, including weaker associations

In the reported runs, the conservative setting produced fewer recommendations. The exploratory setting produced more options, but its average lift was below 1. This means many rules in that setting did not show a stronger association than the consequent’s baseline frequency.

9 Evaluation

Recommendation Examples

Selected basket	Suggested course	Support	Confidence	Lift
Greek Hero	Justice	0.2609	66.87%	1.16
Health in Numbers	CS50x	0.1597	54.70%	1.10
Environmental Health	Justice	0.2426	53.50%	0.92
Greek Hero + CS50x	Justice	0.0616	45.24%	0.78
Justice	Greek Hero	0.2609	45.11%	1.16

The first two examples have lift above 1, so the suggested courses occur with those baskets more often than their baseline rates. The other examples have lift below 1. They satisfy the report’s confidence threshold, but they are

weaker choices if the goal is to recommend courses with positive association. This is why the interface displays lift along with confidence.

Manual Review of Ten Rules

The rules below were reviewed using their metrics and course context. A rule can have a high confidence and still provide weak evidence when its lift is below 1.

Rule	Confidence / lift	Category	Reason
Greek Hero → Justice	66.9% / 1.16	Useful	The rule has lift above 1 and a clear connection between the two courses.
Justice → Greek Hero	45.1% / 1.16	Useful	The reverse rule also has lift above 1, though its confidence is lower.
Health in Numbers → CS50x	54.7% / 1.10	Useful	The association is slightly above the CS50x baseline rate.
Environmental Health → Justice	53.5% / 0.92	Weak / uncertain	Justice is less common in these baskets than its baseline rate.
Justice → CS50x	37.6% / 0.76	Obvious / trivial	CS50x is popular overall; lift does not show a positive association.
Greek Hero → CS50x	34.9% / 0.70	Obvious / trivial	The confidence is not explained by a higher-than-baseline association.
Health in Numbers → Environmental Health	40.8% / 0.90	Weak / uncertain	The lift is below 1, so the pair is less common than the baseline suggests.
Environmental Health → Health in Numbers	26.3% / 0.90	Weak / uncertain	The rule has low confidence and lift below 1.
Greek Hero + CS50x → Justice	45.2% / 0.78	Unexpected	The course combination is less obvious, but lift below 1 means the evidence is weak.
Greek Hero + Justice → CS50x	23.6% / 0.48	Weak / uncertain	The rule has low confidence and a low lift.

10 Limitations

- Apriori treats a basket as an unordered set. It does not show which course a learner took first or whether one course is a prerequisite for another.
- The dataset covers a small course set from 2012–2013. Its patterns may not describe current courses or learners on other platforms.
- Popular courses can appear in many rules. Lift helps identify this, but users still need to consider the course context before acting on a recommendation.
- A new course has no enrollment history, so Apriori cannot recommend it until it appears in enough baskets.
- The data records enrollment, not learning outcomes. It cannot show whether a recommendation improves completion or student performance.

11 Summary

CourseCompass turns learner enrollments into baskets, mines frequent itemsets with Apriori, and uses association rules to suggest courses. The interface shows the evidence for each suggestion and lets users explore different thresholds. The results are useful for examining co-enrollment, but rules with lift below 1 should not be described as strong positive associations.